

Activity - 1
ITA0502 - COMPUTER VISION
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1. Medical Image Enhancement

1. The gray-level transformation that I would recommend is Log Transformation.

2. It is suitable because it enhances the darker region of the x-ray, making soft tissues more visible while keeping bright bone regions clear.

3. Effects on the image

The dark tissues become brighter and more detailed, improve the overall visibility, of the x-ray.

2. Night - Time Surveillance

1. The transformation that would improve the visibility is the Log Transformation

2. It is more suitable than negative transformation because Log Transformation increases the brightness of dark pixels, whereas negative transformation only inverts the image and does not improve visibility.

3. The appearance of the enhanced image will become brighter, with objects like vehicles, people, and the surroundings appearing more visible and easier to recognize.

3. Fingerprint Recognition

1. The gray-level transformation that I would apply is Contrast stretching (piecewise linear transformation)
2. It increases the contrast between, fingerprint, ridges and valleys, making the line ridge details clearer for accurate recognition.
3. If Log Transformation was applied it will brighten the image but would not effectively improve the ridge - to - valley contrast. so fingerprint details may remain unclear.